



An Introduction to Types of Quasi-Experimental Designs

Potentially useful approaches when randomization is not feasible

Editor's note: This is the 24th article in a series on clinical research by nurses. The series is designed to be used as a resource for nurses to understand the concepts and principles essential to research. Each column will present the concepts that underpin evidence-based practice—from research design to data interpretation. To see all the articles in the series, go to <https://links.lww.com/AJN/A204>.

The quasi-experimental design is a research methodology that contains the rigor of a true experimental design (which includes random assignment to at least one control and one experimental/interventional group) and the flexibility of an observational study. This method is often used in social sciences, education, and public health research. Quasi-experimental designs include the posttest-only design with a control group, the one-group pretest–posttest design, and the pretest–posttest design with a control group (see Figure 1). The quasi-experimental method is often used when classic experimental designs are not feasible or ethical, thus bridging the gap between observational studies and true experiments. This article builds on previous articles in this series, introduces the basics of quasi-experimental design, and provides examples of each of the three types.

QUASI-EXPERIMENTAL DESIGN OVERVIEW

Quasi-experimental designs are often utilized when the investigator cannot implement a control group or randomize study groups. If it is not feasible to randomize an intervention or establish a control group, additional factors can be included in the design to strengthen internal validity.¹ As with other research designs, such as randomized controlled trials and cohort studies, a quasi-experimental research design requires that investigators develop eligibility criteria and study aims, and select the appropriate measurement tools to assess outcomes.

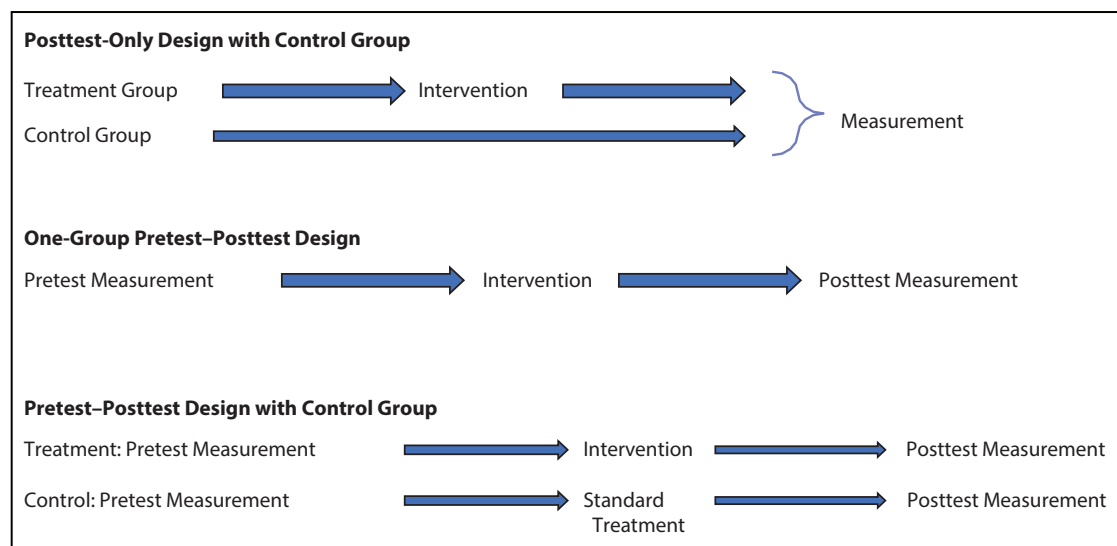
Quasi-experimental studies are often used in real-world settings and can leverage events, such as examining health outcomes following a hurricane or other natural disaster.^{1,2} Suppose a researcher is investigating health outcomes after such an event. In

that case, the study will likely be a quasi-experimental design in which no intervention is given to the study participants. Quasi-experimental designs are also employed for studying behavioral interventions using natural settings (such as a walking initiative developed in a local city). When conducting these studies, the investigator must consider potential sources of bias and threats to validity and select the most appropriate research design.¹ For example, if investigators want to assess stress levels associated with surviving a hurricane, they can't randomize study participants; in this situation, investigators might consider using a quasi-experimental design with the hurricane's geographical location as an eligibility criterion. Other eligibility criteria for this example might include damage or loss of residential dwellings.

For additional information about the quasi-experimental design, investigators can review the 22-item Transparent Reporting of Evaluations with Nonrandomized Designs (TREND) Statement Checklist (available at www.cdc.gov/trendstatement/pdf/trendstatement_trend_checklist.pdf).³ The TREND checklist was developed to improve the reporting quality of nonrandomized behavioral and public health intervention studies. However, the checklist is also excellent for learning about the research design.

POSTTEST-ONLY DESIGN WITH A CONTROL GROUP

There are two groups in the posttest-only design with a control group: an experimental group that receives an intervention and a control (comparison) group that does not. Both groups are evaluated after the intervention. For example, a posttest-only design with a control group might be used to assess the impact on hospital staff of a new hand hygiene intervention to

Figure 1. Examples of Quasi-Experimental Designs

reduce rates of health care–associated infections. For this study, two hospitals could be selected that are similar in size and in level of patient acuity. One hospital would implement the new hand hygiene intervention for its staff while the other would not. After three months, infection rates were collected from both groups. Because working with different institutions might involve restrictive policies or competing interests in implementing new procedures (the need for vendor contracts for certain supplies, for example, or the opposition of hospital management to the product used in the study), posttest-only was the best option for this sample study.

It is essential to acknowledge that because of threats to validity in the above example, such as increased attention to handwashing because of COVID-19, this design does not allow for definitive conclusions regarding causality or efficacy.⁴ Moreover, there is a potential risk of selection bias (flawed procedures used to select participants or institutions, such as poorly defined eligibility criteria) in both groups; including a control group could create a false sense of certainty about the study results. The absence of a pretest, which would involve measuring infection rates before implementing the hand hygiene intervention, hinders the investigators' ability to determine whether any observed disparities in infection rates between the two facilities can be attributed to the intervention or to preexisting discrepancies, such as staffing differences.

ONE GROUP PRETEST-POSTTEST DESIGN

In the one-group pretest–posttest quasi-experimental design, participants are selected based on convenience

and suitability to the study.⁴ In this commonly used design, participants are measured before (pretest) and after (posttest) the intervention. The effect of the intervention is inferred from the difference in pretest and posttest results. For example, a weight-loss study using high-intensity training for 30 minutes five times a week for three months involves weighing study participants before and after the intervention.

Even with the inclusion of a pretest, this design has significant limitations that affect the results.⁴ Pretest results cannot effectively act as a type of control group. Certain events that influence the posttest responses might occur between the pretest and posttest. Posttest scores could be affected by factors such as *historical events* and *maturation*.⁴ An example of a historical event that is considered a threat to internal validity is when an external event unrelated to the study takes place and influences the outcome of the dependent variable. For example, a historical event in the high-intensity training for weight loss study could be a new dietary weight-loss supplement that is promoted on social media and available over the counter. Participants could take the new supplement while completing high-intensity training. Maturation, for this study, poses a risk to internal validity when normal changes stemming from the progression of time affect the outcome of the dependent variable. In other words, high-intensity training may increase muscle mass and body weight.

Additionally, suppose participants have high pretest scores—in other words, close to the desired outcome of lower body weight or lower body mass



index. In that case, the tool may not be adequately sensitive to detect progress, or scores may gravitate toward the mean.⁴ In clinical research, gravitating toward the mean or *regression to the mean* refers to the statistical event in which extreme initial measurements, unusually high or low, tend to move closer to the average in subsequent measurements. Regression to the mean can cause misleading interpretations in research results, especially if not adequately accounted for during the study design and analysis. Incorporating a comparison or control group into this design would enhance the investigator's capacity to draw causal inferences.

Quasi-experimental designs are an alternative option when randomized controlled clinical trials are not feasible or ethical.

PRETEST–POSTTEST DESIGN WITH A CONTROL GROUP

The pretest–posttest design with a control group is another widely used quasi-experimental design. Here, the researcher selects one group to receive the treatment and another with similar characteristics to serve as the control group. Both groups complete a pretest, after which the treatment group receives the intervention; both groups then complete a posttest. It is ideal if the groups' mean scores on the pretest are similar ($P > 0.05$). Additionally, the investigator compares demographic characteristics and other variables such as disease status or time since diagnosis that could potentially influence posttest scores. By ensuring similarity between the treatment and control groups, any differences in posttest scores can be attributed to the intervention.

As an illustration of this design, investigators recruited older adults from two senior centers to assess the impact of an app-based game on the memory of healthy, ambulatory adults ages 75 and above. Participants from Senior Center A were provided with the app-based game and were asked to attend the senior center five days a week for a month, dedicating 30 minutes a day to playing the game and participating in their usual activities. Meanwhile, participants from Senior Center B also attended the senior center five days a week for a month and engaged in their usual activities, such as crafting, dancing, chair yoga, and board games.

Both groups underwent memory tests before and after the 30-day period to measure the efficacy of the app-based game.

This example reveals limitations in this study design's impact on internal validity. One of the main weaknesses is that participants are not randomized into the treatment and control groups. Therefore, any differences observed in the posttest scores of the treatment group may be attributed to an unmeasured confounding variable (a third variable that affects the independent and dependent variables, leading to a distorted association). Additionally, external events or unrelated changes between the pretest and posttest may positively affect the dependent variable of the control group or reduce positive changes in the dependent variable of the treatment group. Potential issues that could influence the outcomes in this example study include some participants in the control group using memory-enhancing nutritional supplements or other memory-based games.

CONCLUSION

In conclusion, although quasi-experimental research designs have limitations, these designs are an alternative option when randomized controlled clinical trials are not feasible or ethical. When choosing a research design, investigators should consider the internal and external validity (the generalizability of the results) trade-offs. Despite the potential shortcomings, quasi-experimental designs can yield valuable insights into causal relationships in real-world settings, provided steps are taken to minimize potential confounders and biases. ▼

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